

Linear Algebra - Question Bank

Course: Linear Algebra | B.Sc Mathematics | Semester 2 | Year 1

1. A matrix with the same number of rows and columns is called:

- a) square
- b) rectangular
- c) diagonal
- d) null

Answer: a

2. The determinant of a 2 by 2 identity matrix is:

- a) 0
- b) 1
- c) 2
- d) -1

Answer: b

3. A vector space must always contain the:

- a) zero vector
- b) unit vector
- c) standard basis
- d) dimension

Answer: a

4. Two vectors are linearly dependent if one is:

- a) a scalar multiple of the other
- b) perpendicular to the other
- c) equal to zero
- d) orthogonal

Answer: a

5. The rank of a matrix equals the number of:

- a) linearly independent rows
- b) total rows
- c) total columns
- d) zero entries

Answer: a

6. The eigenvalues of an identity matrix are:

- a) 0
- b) 1
- c) -1
- d) 2

Answer: b

7. Matrix multiplication is:

- a) commutative
- b) not commutative in general
- c) always equal to addition
- d) undefined for square matrices

Answer: b

8. How many entries does a 3 by 3 matrix have?

- a) 3
- b) 6
- c) 9
- d) 12

Answer: c

9. The transpose of a row vector is a:

- a) column vector
- b) scalar
- c) diagonal matrix
- d) zero matrix

Answer: a

10. A linear system has infinitely many solutions when the coefficient matrix is:

- a) singular
- b) invertible
- c) identity
- d) diagonal

Answer: a

11. Explain what eigenvalues and eigenvectors are.

(Write your answer in the space provided.)

12. State the conditions for a set of vectors to form a basis.

(Write your answer in the space provided.)